

#Project Information

- Project: CleanIT
- Repository: https://github.com/AksilDev/IT342_CleanIT_G4_Macansantos.git
- Branch: vertical-slice-regression-refactor
- Date: 2026-05-09
- Platforms covered: Spring Boot backend, React/Vite web frontend, Android Kotlin mobile app

#Refactoring Summary

The project was refactored from technical-layer folders into vertical feature slices while keeping shared infrastructure separate. Existing API routes, screen behavior, layout resources, database entities, and application entry points were preserved.

Backend slices:

- 'features/auth': authentication controllers, DTOs, auth service, registration validators, auth strategies, user event observers, user factory
- 'features/booking': booking controllers, booking service, booking status DTOs, booking entities, booking repositories, booking enums, booking exception
- 'features/catalog': service catalog and add-on controller, entities, and repositories
- 'features/users': user/admin controllers, user entity, technician settings entity, repositories
- 'shared/config', 'shared/exception', 'shared/storage': security, JWT, application config, global exception handling, Supabase storage

Web slices:

- 'features/auth': login, registration, role selection, OAuth callback
- 'features/booking': booking creation and booking details modal
- 'features/dashboard': client, technician, and admin dashboards
- 'shared/api': Axios and Supabase clients

Mobile slices:

- 'features/auth': login, registration, auth request/response models
- 'features/booking': booking list/detail/create screens, booking adapters, booking request/response models
- 'features/catalog': service list, service/add-on/technician models and adapters
- 'features/dashboard': admin and technician dashboards plus dashboard models
- 'features/home': home and profile screens
- 'shared/network', 'shared/navigation': Retrofit API client and role navigation

#Updated Project Structure

```text

backend/src/main/java/com/G4/backend/

BackendApplication.java

features/

auth/

booking/

catalog/

users/

shared/

config/

exception/

storage/

web/src/

App.tsx

features/

auth/

booking/

dashboard/

shared/api/

mobile/app/src/main/java/edu/cit/macansantos/cleanit/

CleanITApplication.kt

MainActivity.kt

features/

auth/

booking/

catalog/

dashboard/

home/

shared/

navigation/

network/

```

#Functional Requirements Coverage

| Area | Functional requirement | Covered by |

| --- | --- | --- |

| Authentication | Email/password login, registration, JWT session, Google OAuth completion, role routing | Backend auth APIs, web auth screens, mobile auth screens, mobile role navigator |

| Role access | Client, technician, and admin users navigate to role-specific dashboards |

Web routes, mobile 'RoleNavigator', backend security/JWT filter |
Service catalog	List available services, add-ons, technicians, and service add-on compatibility	Backend catalog endpoints, web booking flow, mobile service and booking screens
Booking creation	Client creates bookings with service, device, add-ons, date, time, address, technician assignment	Backend booking service/controller, web booking screen, mobile create booking
Booking management	Client views bookings, details, cancellation, rescheduling	Backend booking APIs, web dashboard/modal, mobile booking list/detail
Technician workflow	Pending/assigned bookings, accept/reject, status updates, checklist, required before/after photos	Backend technician booking APIs, web technician dashboard, mobile technician dashboard
Admin workflow	View dashboard statistics, bookings, user verification, cancellation/status oversight	Backend admin/user APIs, web admin dashboard, mobile admin dashboard
Data integrity	Booking child records are preserved during normal operations and removed correctly when deleting bookings	Backend cascade-delete and preservation tests
Regression safety	Application starts and compiles after vertical slice package moves	Backend tests, web production build, mobile Kotlin compile/unit test

#Test Plan

Test ID	Requirement	Test procedure	Automated evidence
TP-01	Backend application context	Run backend Spring Boot tests and verify application context loads	'backend/target/surefire-reports/com.G4.backend.BackendApplicationTests.txt'
TP-02	Booking checklist initialization	Run checklist initialization regression tests	'ChecklistInitializationBugTest', 'ChecklistPreservationPropertyTest' reports
TP-03	Booking cascade deletion	Delete bookings with photos, checklist rows, add-ons, and all child types	'BookingCascadeDeleteBugTest' report
TP-04	Booking non-deletion behavior	Create/query/update child records and delete child records independently	'BookingCascadeDeletePreservationPropertyTest' report
TP-05	Web route/build integrity	Run Vite production build	'web/dist/index.html', 'web/dist/assets/*'
TP-06	Mobile Kotlin/package integrity	Compile Android debug Kotlin sources	Gradle ':app:compileDebugKotlin' successful output
TP-07	Mobile unit tests	Run Android debug unit tests	'mobile/app/build/test-results/testDebugUnitTest/TEST-edu.cit.macansantos.cleanit.ExampleUnitTest.xml'
TP-08	Manual smoke procedures	Login/register, role dashboard navigation, create booking, update booking, admin verification	Manual procedures listed below

#Manual Regression Steps

1. Register a client, technician, and admin account.
2. Log in as each role and verify redirection to the correct dashboard.
3. As a client, open services, choose a service/add-ons, create a booking, and verify it appears in the client dashboard.
4. As a technician, view assigned bookings, accept a booking, start service, complete checklist, upload before/after photos, and complete service.
5. As a client, open booking details and verify status, photos, schedule, and reschedule/cancel controls.
6. As an admin, verify dashboard statistics, booking list, user verification controls, and booking detail modal.
7. Confirm invalid states are blocked: incompatible add-ons, completing service without checklist/photos, unverified users booking or accepting work.

#Automated Test Evidence

Commands executed on 2026-05-09:

Command	Result
---	---
'backend/.mvnw.cmd -DskipTests compile'	Passed
'backend/.mvnw.cmd test'	Passed: 18 tests, 0 failures, 0 errors
'web/npm run build'	Passed: Vite production build created 'web/dist'
'mobile/gradlew.bat :app:compileDebugKotlin'	Passed
'mobile/gradlew.bat :app:testDebugUnitTest'	Passed

Evidence files:

- Backend Surefire XML/text reports: 'backend/target/surefire-reports/'
- Mobile unit test XML report: 'mobile/app/build/test-results/testDebugUnitTest/'
- Web production build artifacts: 'web/dist/'

#Regression Test Results

Area	Result	Notes
---	---	---
Backend compile	Passed	All moved packages compile under feature slices
Backend tests	Passed	18 tests passed after cascade-delete fix
Web build	Passed	Build completed; Vite reported only existing chunk-size/module-type warnings
Mobile compile	Passed	Kotlin compile passed with existing deprecation/unused-variable warnings
Mobile unit tests	Passed	Debug unit test task completed successfully

#Issues Found

| Issue | Severity | Status | Evidence |

| --- | --- | --- | --- |

| Booking deletion did not reliably remove checklist/add-on child rows when child collections were not attached to the in-memory booking aggregate | High | Fixed | Initial 'backend/.mvnw.cmd test' failed 3 cascade-delete cases |
| First mobile unit-test run could not access Gradle wrapper lock under user '.gradle' directory | Environment | Resolved | Reran with permission; test task passed |

#Fixes Applied

- Refactored backend, web, and mobile code into feature-oriented packages/folders.
- Updated backend package declarations and imports for feature slices.
- Updated web route imports and API-client paths.
- Updated Android package declarations, manifest activity names, resource imports, cross-feature Activity imports, Retrofit model imports, and role navigation imports.
- Moved mobile booking models out of the catalog model file into the booking slice.
- Fixed booking deletion by making 'BookingRepository.delete(Booking)' explicitly delete booking photos, checklist rows, and add-on join rows before deleting the booking row.

#Submission Notes

- Branch to push: 'vertical-slice-regression-refactor'
- Repository link: https://github.com/AksilDev/IT342_CleanIT_G4_Macansantos.git
- PDF filename: 'FullRegressionReport_CleanIT.pdf'
- Supporting automated evidence is available in the build/test report folders listed above.